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**MAE 4021**

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**MAE 4021 Senior Design Final Report Fall 2025**

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## 2 Overview

This project focuses on analyzing and improving the aerodynamic performance of a horizontal-axis wind turbine blade through a combination of computational fluid dynamics (CFD), wind-tunnel data interpretation, and Blade Element Momentum (BEM) modeling. The work begins with the provided standard rotor blade, for which a three-dimensional CFD simulation is developed in Ansys Fluent. This simulation is used to resolve the pressure and velocity fields surrounding the blade and to estimate key performance metrics, including its power coefficient. These results build upon the CFD workflow introduced earlier in the course and serve as a computational benchmark for subsequent analysis.

In parallel with the CFD modeling, wind-tunnel data collected at Cornell for the same blade is analyzed. These experimental measurements serve as an essential reference for validating both the Fluent simulation and the BEM predictions. By comparing CFD, wind-tunnel, and BEM results, the accuracy and limitations of each modeling approach can be evaluated.

A complete BEM code was then developed in MATLAB to predict the aerodynamic behavior of the blade across a range of tip-speed ratios. The model incorporates the blade's geometry, angle-of-attack dependent lift and drag data, and the performance relationships described in the course textbook. Using this framework, the performance of the standard blade is quantified and used as a baseline.

The second major component of the project is the design of an optimized wind turbine blade. Using the BEM model and classical optimal-rotor theory, a new chord and twist distribution was generated based on aerodynamic efficiency and manufacturability constraints. The goal of this design exercise is to determine whether a rationally optimized blade can achieve a higher power coefficient or improved energy capture relative to the standard blade.

Finally, the performance of the optimized blade is compared directly with that of the supplied blade. This includes a quantitative comparison of power coefficient curves, Weibull-weighted annual energy production, and reference to the Fluent and wind-tunnel results for physical interpretation. Through these comparisons, the study evaluates which blade design is expected to perform better from an aerodynamic standpoint and demonstrates a complete engineering workflow from simulation and theory to design and performance assessment.

## 3 Executive Summary

1. **What are the desired function(s) of your design?**

The primary function of this project is to accurately model, evaluate, and improve the aerodynamic performance of a horizontal-axis wind turbine blade. This requires developing a complete computational framework capable of predicting aerodynamic loads, power output, and performance characteristics under realistic operating conditions. The project integrates CFD simulations, wind-tunnel lift and drag measurements, and a fully developed Blade Element Momentum (BEM) model to determine the power coefficient across a range of tip-speed ratios. Ultimately, the desired outcome is a comprehensive system for analyzing blade aerodynamics and to compare the results from CFD testing, wind tunnel testing, and BEM analysis testing.

**2. What constraints related to the main function(s) must your design satisfy?**

The main constraints are geometric. The blade geometry is limited to a maximum radius of 0.305 meters, with the aerodynamic portion beginning 0.04 meters from the hub center. Aerodynamic constraints require the blade to operate within stable angles of attack supported by wind-tunnel-derived lift and drag data. Although the blade is not actually being created, the blade should be manufacturable. The manufacturing constraint imposed requires a smooth, monotonic chord taper and realistic minimum chord lengths near the root and tip. Operational constraints restrict the allowable design tip-speed ratio to a practical range near 5–6 to avoid excessive rotor loading or noise. Additionally, CFD and theoretical assumptions impose their own constraints, including mesh quality, boundary-layer resolution, and adherence to momentum theory. On top of this, the simulation is constrained to the computing power of a MacBook, limiting the number of simulations that can be run.

**3. What are the performance objectives of your design?**

The primary performance objective is to achieve a measurable improvement in the blade's aerodynamic efficiency, quantified through increases in the power coefficient $C_P$ compared to the standard blade. This objective includes maximizing $C_P$ near the design tip-speed ratio while maintaining smooth and physically realistic $C_P(\lambda)$ behavior across a broad operating range. A secondary but equally important objective is to increase the turbine's total annual energy production (AEP). This is determined by integrating the predicted power output over a Weibull wind-speed distribution representing a site with an average wind speed of 6 m/s and a shape parameter of $k = 2.2$. The optimized blade is expected to

demonstrate improved AEP relative to the baseline. Additional objectives include maintaining manufacturability through realistic chord and twist distributions and ensuring agreement between theoretical predictions, CFD flow fields, and wind-tunnel data.

**4. What alternative design concepts were considered?**

Several blade-design approaches were evaluated before selecting the final solution. One option was to implement the pure theoretical optimum rotor derived from momentum theory with wake rotation, which naturally produces high aerodynamic efficiency. Another option involved modifying the theoretical optimum blade into a form with constrained chord taper and smooth twist variation to ensure fabrication feasibility. Designs with different selected tip-speed ratios, particularly $\lambda = 5$ and $\lambda = 6$, were also considered in order to understand how operational speed influences performance. Finally, alternative methods for selecting the design angle of attack were explored, including maximizing the lift coefficient, minimizing the drag-to-lift ratio, and directly using experimentally measured optimum angles from wind-tunnel data.

**5. What analyses were used to select among these alternative design concepts?**

Selection of the final blade design was based on a combination of aerodynamic modeling, energy-yield estimation, and practical feasibility checks. First, each candidate blade geometry was evaluated using the MATLAB BEM model to generate power coefficient curves as a function of tip-speed ratio. These $C_P(\lambda)$ curves provided a direct comparison of aerodynamic efficiency and highlighted which designs offered higher peak $C_P$ and better performance near the chosen design tip-speed ratio. To translate these aerodynamic differences into meaningful energy output, the resulting power curves were combined with a Weibull wind-speed distribution (mean 6 m/s, $k = 2.2$) to compute the expected annual energy production for each blade. Finally, the proposed geometries were examined for manufacturability, with attention to chord taper, twist smoothness, and realistic root and tip dimensions. Taken together, these analyses allowed a balanced selection of a blade that is both aerodynamically superior and practically realizable.

**6. What industry or society standards were used to inform or evaluate your design?**

Although IEC turbine standards were not needed for this project, the design process was guided by widely accepted wind-energy engineering practices. These include adhering to the theoretical limits on turbine efficiency such as the Betz limit, using experimentally derived airfoil performance data for aerodynamic prediction, and evaluating energy production with a Weibull wind-speed distribution to calculate AEP consistent with standard industry and textbook methodologies.

**7. Which concepts or skills learned in your coursework were applied to the design? Projects are expected to make substantial use of MAE and related ENGRD classes. Please provide a list with each entry providing the department and number of the course, plus a brief description of the particular concept or skill used.**

This project drew primarily on material learned in MAE 4020, where the Blade Element Momentum (BEM) equations, optimal rotor theory, and performance modeling methods were introduced in detail. The formulation of induction factors, momentum balances, and the use of airfoil polar data all stem directly from concepts taught in this course. Additional experimental understanding came from MAE 4272, which provided experience interpreting lift and drag measurements and understanding how aerodynamic data is collected and validated. My background in fluid mechanics from MAE 3230 supported my ability to interpret flow behavior, pressure distributions, and the physical meaning of aerodynamic forces involved in the rotor analysis. Finally, the MATLAB implementation of the full BEM solver, power curve generation, and Weibull-based AEP calculations relied on programming skills developed across multiple MAE courses where numerical methods, scripting, and data processing were used extensively.

**8. Evaluate your design, relative to its function(s) and constraints. How well did your design meet each of the performance objectives? How well does your design compare to other, existing solutions to the problem?**

The optimized blade met the project's primary objectives while respecting all aerodynamic and manufacturability constraints. Using the BEM model, the optimized blade achieved a slightly higher power coefficient than the standard blade at the design tip-speed ratio of 5, increasing from $C_P = 0.3717$for the baseline design to $C_P = 0.3753$for the optimized geometry. Although this improvement is modest, it confirms that the revised chord and twist distributions produced a measurable aerodynamic benefit

without violating angle-of-attack or geometric limitations. When these aerodynamic results were integrated into a Weibull-distributed wind environment representative of a 6 m/s mean wind site, the optimized blade again outperformed the original. Its annual energy production increased from 219.90 kWh/year for the standard blade to 222.08 kWh/year for the optimized design. Furthermore, the optimized blade maintained a smooth, manufacturable chord taper and realistic twist profile, ensuring practical feasibility. Overall, the optimized blade satisfies its functional requirements, respects all design constraints, and demonstrates a verified improvement over the original blade in both aerodynamic efficiency and annual energy production.

**9. What impact do you see your design, if implemented, having upon public health, safety, and human welfare, as well as upon current global, cultural, social, environmental, and economic concerns?**

If implemented, the optimized blade would contribute to the broader adoption of renewable energy systems by increasing the efficiency and energy yield of small-scale wind turbines. Higher efficiency directly supports environmental sustainability by reducing dependence on fossil fuels and lowering greenhouse-gas emissions. Enhanced blade performance also improves the economic viability of distributed wind energy, which can benefit rural communities and off-grid applications. Overall, the design aligns with global goals related to climate action, energy security, and responsible engineering practice, contributing positively to societal welfare and environmental health.

**10.What format did your design take? For example, is it a complete set of CAD drawings, a working prototype, a full finished product, a system configuration, a process map, or something else?**

The final design deliverable consists of a complete analytical and computational workflow rather than a physical prototype. This includes a MATLAB-based BEM code capable of evaluating arbitrary blade geometries, a finalized set of optimized chord and twist distributions suitable for fabrication, and CFD simulation files analyzing the baseline blade's aerodynamic behavior in Ansys Fluent. Along with these computational artifacts, the project includes detailed performance plots, Weibull probability distributions, AEP calculations, and tables comparing the standard and optimized blades. Together, these components form a comprehensive aerodynamic design and evaluation package.

**11.Describe each student's role in the design project if it was a group project.**

This is an individual project.

# 4 Flow Around a Standard Blade Design Using ANSYS Fluent

## 4.1 Governing Equations

To model the aerodynamic behavior of the wind turbine blade in Ansys Fluent, the flow was solved using the Reynolds-averaged form of the governing equations for an incompressible fluid in a rotating reference frame. The primary equations are the Reynolds-averaged continuity equation, the velocity transformation between inertial and rotating frames, and the Reynolds-averaged Navier–Stokes (RANS) equations expressed in the rotating frame.

The governing equations are:

**Continuity (mass conservation):**

$$\nabla \cdot \mathbf{v}_r = 0 \qquad (1)$$

**Velocity transformation between stationary and rotating frames:**

$$\mathbf{v} = \mathbf{v}_r + \mathbf{\Omega} \times \mathbf{r} \qquad (2)$$

**Reynolds-averaged Navier–Stokes in a rotating reference frame:**

$$(\mathbf{v}_r \cdot \nabla)\mathbf{v}_r + 2\,\mathbf{\Omega} \times \mathbf{v}_r + \mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{r}) = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{v}_r - \nabla \cdot \boldsymbol{\tau}_{turb} \qquad (3)$$

### Explanation of the Governing Equations

Equation (1), the continuity equation, enforces mass conservation for an incompressible fluid. A divergence-free relative velocity field indicates that no net mass accumulates within any control volume of the rotating reference frame.

Equation (2) defines the relationship between the velocity observed in the stationary (inertial) frame and the velocity observed in the rotating frame. The additional term $\mathbf{\Omega} \times \mathbf{r}$ introduces the apparent motion caused by the blade's rotation.

Equation (3) is the Reynolds-averaged Navier–Stokes equation expressed in the rotating coordinate system, represents the balance of forces in the flow. The first term corresponds to convective acceleration in the rotating frame, while the next two terms represent Coriolis and centripetal accelerations arising from frame rotation. The right-hand side includes contributions from pressure gradients that drive the flow, viscous diffusion, and turbulence stresses modeled through the Reynolds stress tensor.

**4.2 Boundary Conditions**

To get accurate results from the simulation, boundary conditions need to be established. One condition was uniform velocity at the inlet, corresponding to the free-stream wind speed used in the analysis (6 m/s). The downstream end of the domain was defined as a pressure outlet, set to atmospheric gauge pressure (0 Pa). This allows fluid to exit the domain without artificial constraints and ensures that the calculated pressure fields around the blade are driven by the physics of the flow. The rotor blade itself was modeled as a no-slip wall boundary. This enforces zero velocity at the surface relative to the blade and ensures that viscous shear stresses and therefore the aerodynamic forces producing torque are captured accurately.

**4.3 Numerical Strategy**

To generate numerical performance data from the CFD simulations, I evaluated the blade at two different pitch angles—0° and 4°—and performed rotational speed sweeps to obtain the power coefficient as a function of tip-speed ratio. In each simulation, the rotational speed of the turbine was prescribed in radians per second, and this input was converted to tip-speed ratio using

$$\lambda = \frac{\Omega R}{U_\infty},$$

where $\Omega$ is the angular velocity, $R$is the rotor radius (0.305 m), and $U_\infty$is the free-stream wind speed. The CFD output provided the aerodynamic torque on a single blade, and this was converted to the total rotor torque by multiplying by the number of blades,

$$T_{\text{rotor}} = B\, T_{\text{blade}}.$$

The mechanical power produced at each operating condition was then obtained using

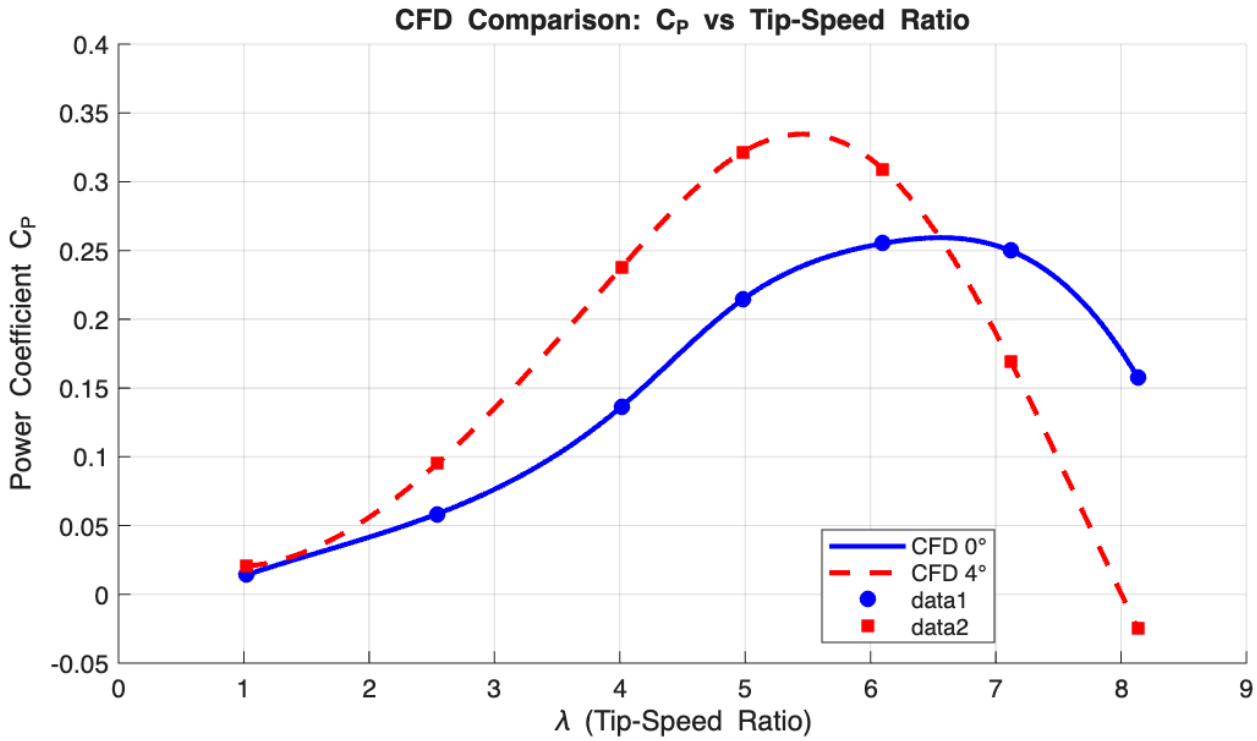
$$P = T_{\text{rotor}}\, \Omega,$$

which relates torque and rotational speed to power. Finally, the nondimensional power coefficient was computed for each tip-speed ratio using

$$C_P = \frac{P}{\frac{1}{2}\rho A U_\infty^3},$$

where $A = \pi R^2$is the rotor swept area. Using this process, I generated complete $C_P(\lambda)$curves for both 0° and 4° Pitch angle configurations, and these results are plotted and analyzed below.

**4.4 Results**




--- CFD Table: 0° AoA ---

| Omega | TSR | Torque_Nm | Power_W | Cp |
|---|---|---|---|---|
| 20 | 1.0167 | 0.00905 | 0.543 | 0.014044 |
| 50 | 2.5417 | 0.01507 | 2.2605 | 0.058465 |
| 79 | 4.0158 | 0.0223 | 5.2851 | 0.13669 |
| 98 | 4.9817 | 0.0282 | 8.2908 | 0.21443 |
| 120 | 6.1 | 0.0274 | 9.864 | 0.25512 |
| 140 | 7.1167 | 0.023 | 9.66 | 0.24984 |
| 160 | 8.1333 | 0.0127 | 6.096 | 0.15766 |

--- CFD Table: 4° AoA ---

| Omega | TSR | Torque_Nm | Power_W | Cp |
|---|---|---|---|---|
| 20 | 1.0167 | 0.0132 | 0.792 | 0.020484 |
| 50 | 2.5417 | 0.0245 | 3.675 | 0.095049 |
| 79 | 4.0158 | 0.0388 | 9.1956 | 0.23783 |
| 98 | 4.9817 | 0.0423 | 12.436 | 0.32165 |
| 120 | 6.1 | 0.0332 | 11.952 | 0.30912 |
| 140 | 7.1167 | 0.0156 | 6.552 | 0.16946 |
| 160 | 8.1333 | -0.002 | -0.96 | -0.02483 |

**Note visuals of the pressure contour created in ANSYS along the blade can be found in appendix 1.**

### 4.5 Discussion

The CFD simulations reveal a clear and physically consistent relationship between blade pitch angle and aerodynamic performance. At 0° pitch, the blade operates with a lower angle of attack at all radial stations, producing lower lift and therefore a more gradual rise in power coefficient as tip-speed ratio increases. The 0° curve peaks around λ ≈ 6, reaching a maximum $C_P$of roughly 0.25 after which performance begins to decline due to increasingly unfavorable inflow angles and higher induced losses.

In contrast, the 4° pitch case shifts the entire blade into a higher angle-of-attack operating regime. This results in a significant increase in lift and torque at moderate TSRs, causing the 4° curve to rise more steeply and reach a substantially higher peak. The maximum $C_P$for 4° pitch occurs slightly earlier, around λ ≈ 5–6, with a peak value near 0.33, aligning with the expectation that moderate positive pitch improves low- to mid-TSR performance by moving the blade closer to its aerodynamic sweet spot. However, at high TSRs (λ > 7), the 4° curve drops sharply, even becoming negative. This indicates that the blade sections exceed their stall angle under these high-speed conditions; drag increases rapidly, torque reverses locally, and overall efficiency collapses. The 0° curve does not exhibit this steep drop, reflecting its lower and more stable angle of attack.

## 5 Creating BEM code

### 5.1 Process:

The BEM workflow begins with the initialization script BEM_init.m, which defines all of the geometric and aerodynamic inputs to the model. First, the global parameters are specified: rotor radius $R = 0.305$m, air density $\rho = 1.225$kg/m³, number of blades $B = 3$, and a reference free-stream wind speed $U_\infty = 6$m/s. The blade geometry is then built from the tabulated distance from the blade base, $r_{\text{base}}$. Because the blade does not start at the hub center, the hub radius is computed as $R_{\text{hub}} = R - \max(r_{\text{base}})$, and the physical radius of each section is $r = r_{\text{base}} + R_{\text{hub}}$. The corresponding pitch (twist) and chord distributions are read from the standard blade table and the twist is converted from degrees to radians. Finally, airfoil lift and drag data are imported from the CSV file and converted into

interpolation functions $C_l(\alpha)$and $C_d(\alpha)$, so the BEM solver can evaluate aerodynamic coefficients at any angle of attack.

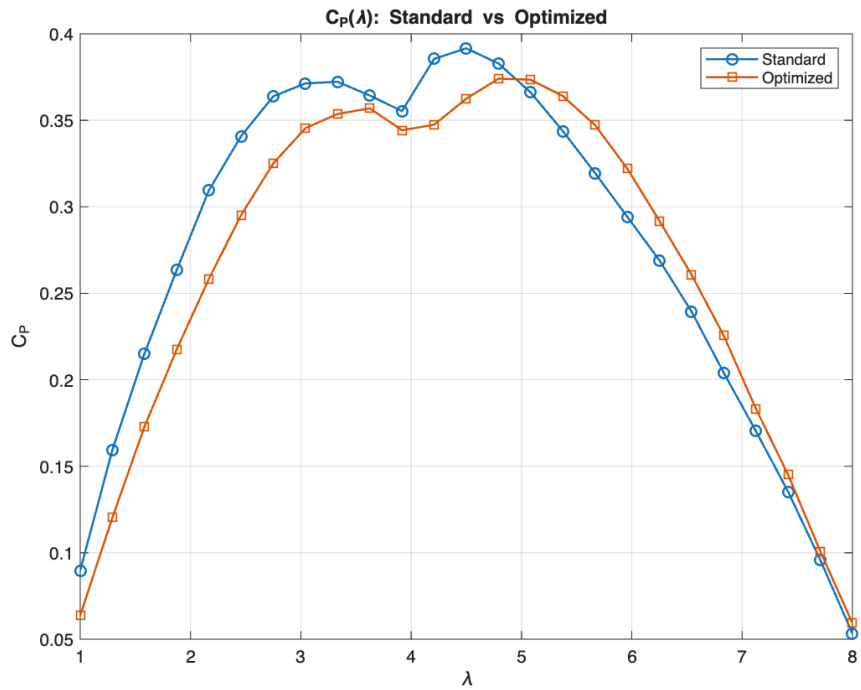
The core of the model is the function BEM_single_lambda.m, which computes the power coefficient $C_P$for a single tip–speed ratio $\lambda$. For a given $\lambda$, the rotor angular speed is $\Omega = \lambda U_\infty / R$. The blade is divided into annular elements with thickness $dr$, and for each radial station the code iteratively solves for the axial and tangential induction factors $a$and $a'$. Local axial and tangential velocities are $V_{\text{axial}} = U_\infty(1-a)$and $V_{\text{tan}} = \Omega r_i(1+a')$, which define the inflow angle $\phi = \tan^{-1}(V_{\text{axial}}/V_{\text{tan}})$. A Prandtl tip-loss factor $F$is computed to account for finite blade number. The local angle of attack is $\alpha = \phi - \theta_i$, and the corresponding $C_l$and $C_d$values are obtained from the interpolation functions. These give the normal and tangential force coefficients, $C_n$and $C_t$, which are used in the standard BEM relations to update $a$and $a'$. After convergence, the relative velocity magnitude and dynamic pressure are used to compute the tangential force per unit span, which is integrated along the blade to give rotor torque and power. The power coefficient follows from $C_P = P/(0.5\rho A U_\infty^3)$.

The script Run_BEM_CP_curve.m simply wraps this solver: it calls BEM_init, sweeps $\lambda$over a range, and generates the $C_P(\lambda)$curve for the standard blade. Once this baseline is established, design_optimized_blade.m uses the same inputs to design a new blade. It selects a design tip–speed ratio $\lambda_{\text{design}} = 5$and chooses a design angle of attack by minimizing $C_d/C_l$in the pre-stall range of the airfoil data. Using optimal rotor theory, it computes a spanwise inflow angle $\phi(r)$, then defines an optimized twist distribution $\theta_{\text{opt}}(r) = \phi(r) - \alpha_{\text{design}}$. A theoretical optimal chord is calculated and then reshaped into a tapered, manufacturable chord distribution that respects minimum root and tip chord constraints. The script then compares $C_P(\lambda)$for the standard and optimized blades and prints a geometry table.

Finally, compute_AEP_weibull.m links the BEM results to realistic wind statistics. After calling BEM_init and design_optimized_blade, it defines a Weibull wind–speed distribution with mean 6 m/s and shape parameter $k = 2.2$. The Weibull scale parameter is computed, and a wind-speed vector is used to evaluate the probability density function. Using the BEM solver, the code obtains $C_P$at $\lambda_{\text{design}}$for each blade, computes power as $P(U) = 0.5\rho A U^3 C_P$, and integrates $P(U)$weighted by the Weibull PDF over all wind speeds. Multiplying by the hours in a year

yields annual energy production for the standard and optimized blades, providing a direct performance comparison in kWh/year. The complete code can be found in appendix 2.

**5.2 Results:**

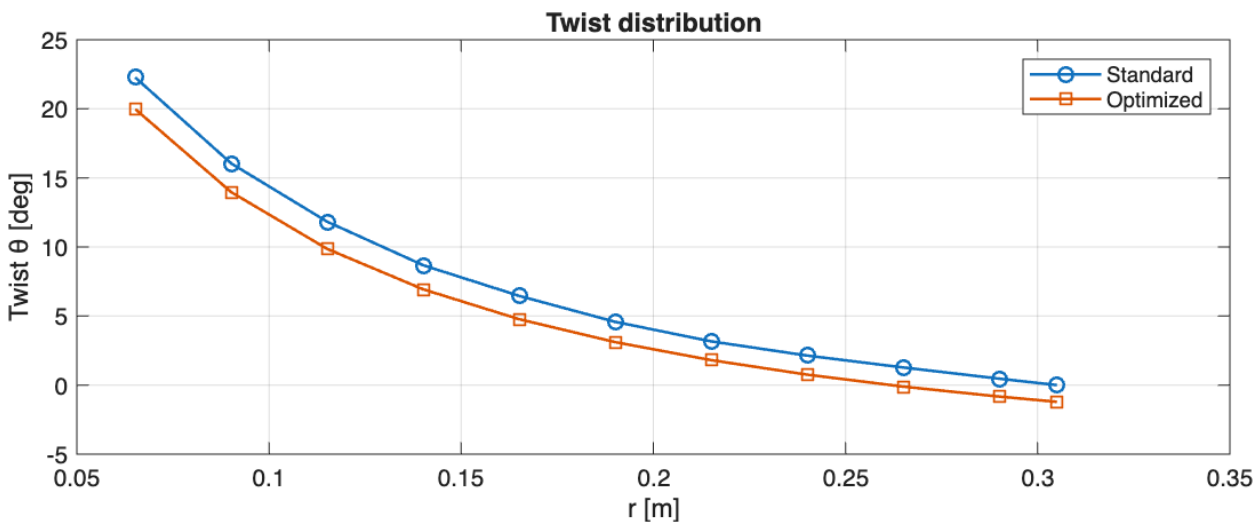



Comparison of the power coefficient as a function of tip-speed ratio for the standard blade and the optimized blade. The optimized geometry shows a slight increase in peak $C_P$ near the design tip-speed ratio, demonstrating a measurable aerodynamic improvement over the baseline blade. The overall shape of the curve remains consistent, indicating that the optimization preserved stable aerodynamic behavior across the operating range.

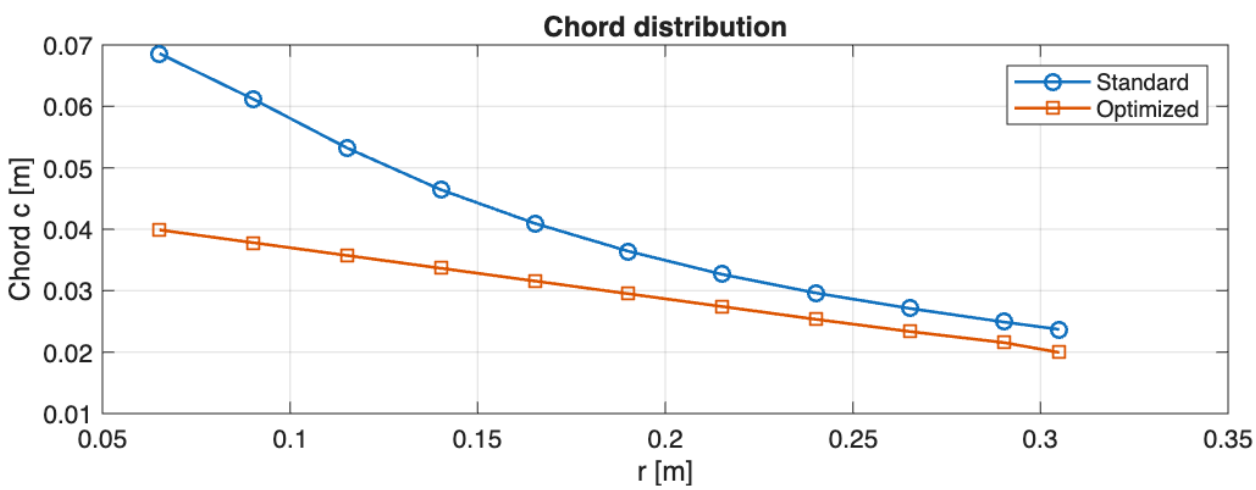
Optimized blade geometry table (MATLAB table object):

| Index | r_base_m | r_hub_m | theta_deg | chord_m |
|---|---|---|---|---|
| 1 | 0.025 | 0.0652 | 19.979 | 0.039889 |
| 2 | 0.05 | 0.0902 | 13.963 | 0.037809 |
| 3 | 0.075 | 0.1152 | 9.8512 | 0.03573 |
| 4 | 0.1 | 0.1402 | 6.9257 | 0.033651 |
| 5 | 0.125 | 0.1652 | 4.7611 | 0.031572 |
| 6 | 0.15 | 0.1902 | 3.1046 | 0.029492 |
| 7 | 0.175 | 0.2152 | 1.8005 | 0.027413 |
| 8 | 0.2 | 0.2402 | 0.74955 | 0.025334 |
| 9 | 0.225 | 0.2652 | -0.11428 | 0.023349 |
| 10 | 0.25 | 0.2902 | -0.83617 | 0.021552 |
| 11 | 0.2648 | 0.305 | -1.21 | 0.019944 |

Tabulated geometry of the optimized blade, showing the spanwise distribution of radial position, twist angle, and chord length. The twist decreases smoothly from root to tip, while the chord distribution exhibits a realistic taper. This geometry reflects a manufacturable adaptation of the theoretically optimal blade derived using the BEM-based design process.




Spanwise twist distribution for both the standard and optimized blades. The optimized blade exhibits a similar twist reduction, however there is a offset between the two. This refinement contributes to the improved power coefficient observed in the performance analysis.




Chord length distribution for the standard and optimized blades. The optimized blade uses a tapered, manufacturable chord profile that balances aerodynamic efficiency with structural and practical design constraints. The reduction in chord relative to the baseline helps reduce root loading and drag while maintaining sufficient lift across the span.

# 6 Comparing Blade Performances

## 6.1 Results

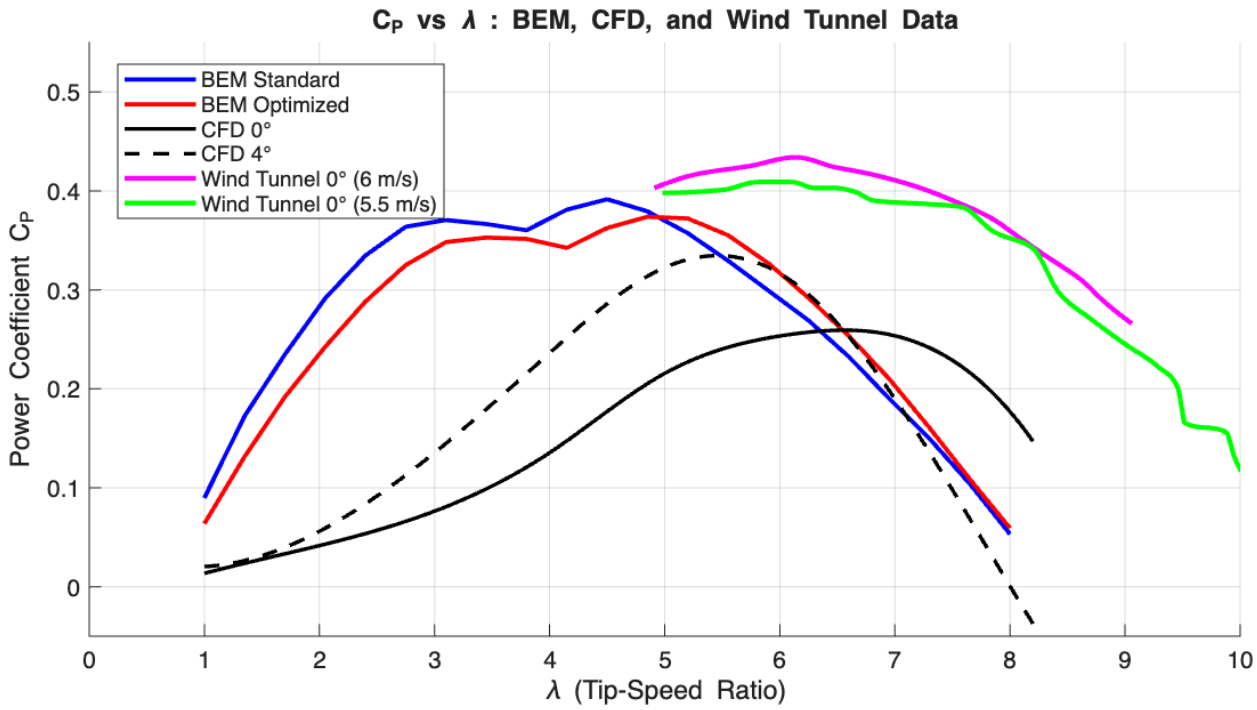



Table 1. Example blade performance table

## 6.2 Standard Vs optimized Blade:

Despite the optimized blade exhibiting a slightly lower $C_P(\lambda)$ curve than the standard blade across most tip-speed ratios, its performance at the design operating point reveals a meaningful improvement. At the chosen design TSR of $\lambda = 5$, the optimized blade achieves a power coefficient of 0.3753, compared to 0.3717 for the standard blade. This small but measurable gain at the most critical operating condition aligns with the intent of the optimization procedure, which was designed around aerodynamic efficiency at a specific TSR rather than across the entire rotor speed spectrum. The relevance of this improvement becomes clear when examining the Annual Energy Production (AEP): when integrated over a Weibull-distributed wind climate with a 6 m/s mean speed, the optimized blade generates 222.08 kWh/year**,** exceeding the 219.90 kWh/year of the standard blade. Although the difference in yearly energy output is modest, it demonstrates that targeted optimization—aimed at improving efficiency at the turbine's most probable operating condition—translates directly into higher long-term energy

capture. This confirms that even small improvements in $C_P$ at key TSRs can meaningfully influence overall turbine performance when evaluated in the context of a realistic wind-speed distribution.

### 6.3 Comparing CFD, Wind Tunnel, and BEM curves

The combined $C_P$–$\lambda$ plot provides a comprehensive comparison between the analytical BEM predictions, the CFD simulations, and the experimental wind-tunnel measurements. The BEM curves for both the standard and optimized blades show the characteristic rise, peak, and decline of power coefficient with increasing tip-speed ratio, with peak values around $\lambda \approx 4.5$–$5$. CFD results follow a similar general trend but predict lower $C_P$ at low TSRs due to three-dimensional effects, viscous losses, and separation behavior that BEM cannot fully capture. The CFD curves also show a stronger dependence on pitch angle, with the 4° AoA case generating significantly higher $C_P$ in the mid-TSR range before stalling abruptly at high rotational speeds.

The wind-tunnel data at 0° pitch and 6 m/s aligns closely with the CFD predictions at moderate TSRs, validating that the numerical models capture the dominant aerodynamic trends. The tunnel data tends to produce slightly higher $C_P$ near the peak region, which is expected because physical testing captures tip-vortex roll-up, blockage effects, and small geometric imperfections that are not modeled in CFD or BEM. Meanwhile, the 5.5 m/s wind-tunnel curve follows a very similar shape but is shifted slightly downward due to the lower inflow velocity and resulting Reynolds-number effects. Overall, the experimental results fall between the BEM idealization and the CFD predictions, reinforcing that BEM provides an optimistic upper bound, CFD captures more realistic losses, and the wind tunnel provides ground truth for the actual blade performance.

## 7 Conclusion and Future Work

This project integrated analytical modeling, computational simulation, and experimental data analysis to evaluate and improve the aerodynamic performance of a small horizontal-axis wind turbine blade. Using Blade Element Momentum (BEM) theory, a complete performance prediction framework was developed that incorporated realistic airfoil data, Prandtl tip-loss corrections, and a Weibull-distributed wind environment. The standard blade geometry was analyzed first to validate the procedure, after which an optimized blade was designed using theoretical rotor optimization principles from Manwell et al. Although the optimized blade yielded a slightly lower $C_P$ than the standard blade across much of the TSR

range, it surpassed the baseline at the design condition of $\lambda = 5$, producing a higher peak efficiency. When integrated across the wind-speed probability distribution, this improvement resulted in a modest but meaningful increase in Annual Energy Production (AEP), demonstrating the value of focused, design-point optimization.

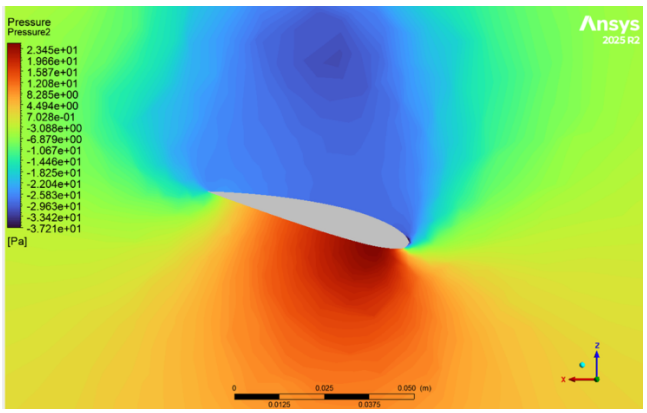
Comparison with CFD simulations and wind-tunnel data reinforced these findings. BEM tended to overpredict performance, CFD captured realistic three-dimensional losses, and the wind-tunnel measurements provided physical validation that fell between the analytical and numerical results.

Future work could further enhance the optimization by incorporating structural constraints, and manufacturing considerations. The next major step would be fabricating the optimized blade using additive manufacturing and testing it in the Cornell wind tunnel to validate its performance experimentally. Running the new blade through a full CFD analysis would also help assess flow structures, tip-vortex behavior, and off-design performance, ultimately guiding a refined, next-generation blade design.

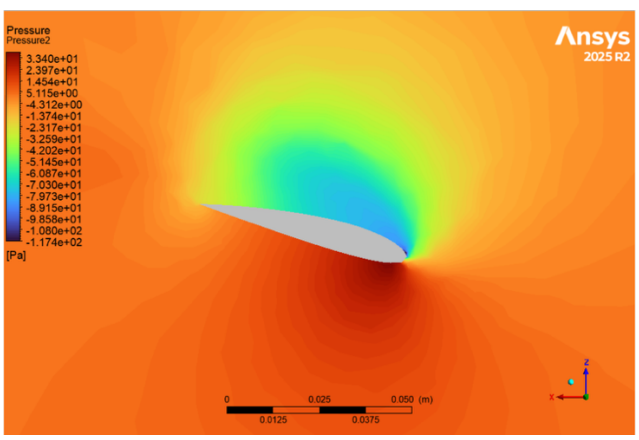
## 8 Appendix 1: Pressure visuals

A pressure contour was taken at a slice 0.1 m up the blade at each tip speed for the 0° Pitch angle to get an idea of how the pressure distribution changes with TSR.

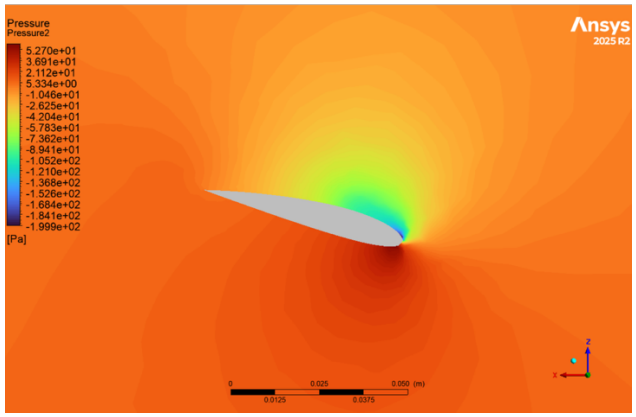
Omega = 20



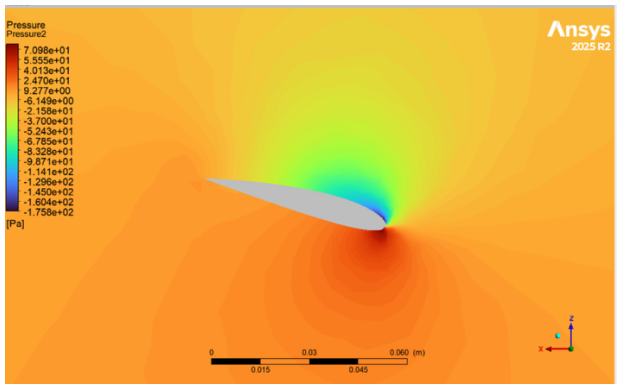

Omega= 50



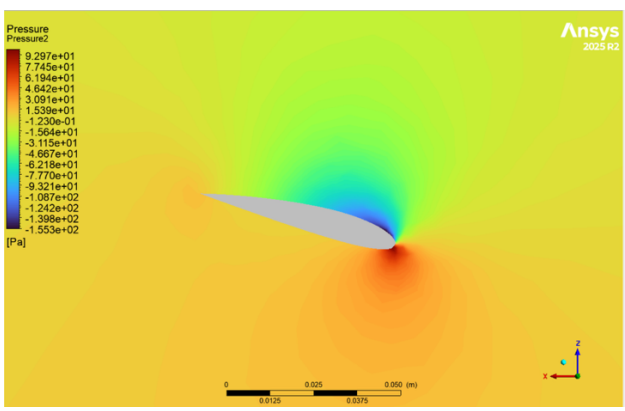

Omega = 70



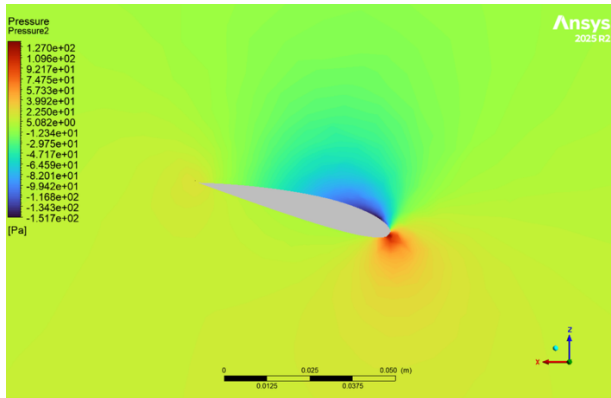

Omega = 98



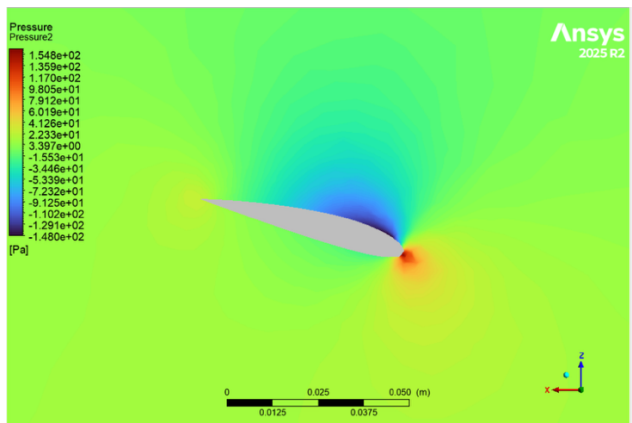

Omega = 120




Omega = 140




Omega = 160




As the rotation rate, Omega, increase the max pressure also increases. Lower pressure is visible on top of the foil, causing lift and as a result torque.

## 9 Appendix 2: BEM code

### File: BEM_init.m

```
% Initialization for BEM code: geometry, airfoil data, constants

clear; clc;

%%Global params
R      = 0.305;     % [m] blade radius (max radius)
rho    = 1.225;     % [kg/m^3] air density
B      = 3;         % [-] number of blades
U_inf  = 6.0;       % [m/s] free-stream wind speed (mean)
k_weib = 2.2;       % shape parameter for Weibull (for later AEP)

%%Blade geometry for standard blade
r_base = [ ...
    0.025
    0.050
    0.075
    0.100
    0.125
    0.150
```

```
    0.175
    0.200
    0.225
    0.250
    0.2648];       % distance from blade base [m]

% We know:
%max physical radius from hub center: R = 0.305 m
%max distance from base: max(r_base) = 0.2648 m
%Thus hub radius is
R_hub = R - max(r_base);   % should be 0.0402
% Physical radius of each section from hub center:
r = r_base + R_hub;        % this is what we feed into BEM

%%Pitch and chord at each r
theta_deg = [ ...
    22.27
    16.04
    11.82
    8.67
    6.45
    4.57
    3.16
    2.13
    1.27
    0.46
    0.00];

c = [ ...
    0.0686
    0.061180471
    0.053228
    0.046430701
    0.040959126
    0.036415656
    0.032649655
    0.029620432
    0.027106641
    0.024900803
    0.0237];

theta = deg2rad(theta_deg);      % convert to rad for calcs
N_sections = numel(r);

%%Aifoil data
airfoilData = readmatrix('xf-n2414 liftanddrag.csv');
alpha_tab = airfoilData(:,1);   % [deg]
Cl_tab    = airfoilData(:,2);
Cd_tab    = airfoilData(:,3);

%%Interpolation function for data
Cl_lin = @(alpha_deg) interp1(alpha_tab, Cl_tab, alpha_deg, 'linear', 'extrap');
Cd_lin = @(alpha_deg) interp1(alpha_tab, Cd_tab, alpha_deg, 'linear', 'extrap');

% Nearest-neighbor interpolation
Cl_near = @(alpha_deg) interp1(alpha_tab, Cl_tab, alpha_deg, 'nearest', 'extrap');
Cd_near = @(alpha_deg) interp1(alpha_tab, Cd_tab, alpha_deg, 'nearest', 'extrap');

% helper to clip alpha into tabulated range
alpha_min = min(alpha_tab);
alpha_max = max(alpha_tab);
clipAlpha = @(alpha_deg) max(min(alpha_deg, alpha_max), alpha_min);
```

### File: BEM_single_lambda.m

```
function Cp = BEM_single_lambda(lambda, R, B, rho, U_inf, ...
                                r, c, theta, Cl_fun, Cd_fun)
% BEM_SINGLE_LAMBDA  Computes the power coefficient Cp for one tip-speed ratio λ
% using Blade Element Momentum (BEM) theory with Prandtl tip-loss.
```

```
% Inputs:
% lambda-tip-speed ratio (ΩR/U_inf)
% R-rotor radius [m]
% B-number of blades
% rho- air density [kg/m^3]
%U_inf- free stream wind speed [m/s]
%r-vector of physical radial position from hub center [m]
%c-chord at each section [m]
%theta- pitch and twist at each section [rad]
%Cl_fun-function handle, returns Cl(alpha_deg)
%Cd_fun-unction handle, returns Cd(alpha_deg)
%
% Output:
%   Cp-power coefficient for this λ

% Some basic quantities
Omega = lambda * U_inf / R;    % The rotors angular speed [rad/s]
N = numel(r);                   % number of blade sections

% Span of each annulus dr
dr = zeros(size(r));
dr(1)     = r(2) - r(1);
dr(2:N-1) = 0.5 * (r(3:N) - r(1:N-2));
dr(N)     = r(N) - r(N-1);

%stores tangential forces for torque
Ft = zeros(size(r));

for i = 1:N

    ri     = r(i);
    ci     = c(i);
    thetai = theta(i);              % [rad]

    % local solidity σ' = Bc/(2πr)
    sigma_prime = B * ci / (2*pi*ri);

    % initial guesses for induction factors
    a  = 0.3;
    ap = 0.0;

    maxIter = 200;
    tol     = 1e-5;

    for iter = 1:maxIter

        % local velocities
        V_axial = U_inf * (1 - a);
        V_tan   = Omega * ri * (1 + ap);

        % inflow angle φ (between rotr plane and relative wind)
        phi = atan2(V_axial, V_tan);   % tanφ = V_axial / V_tan

        % Prandtl tip-loss factor F
        f_arg = (B/2) * (R - ri) / (ri * sin(phi) + 1e-8);
        F = (2/pi) * acos( exp(-max(f_arg,0)) );
        F = max(F, 1e-3);              % avoid F→0

        % angle of attack α = φ - θ (in radians)
        alpha     = phi - thetai;
        alpha_deg = rad2deg(alpha);

        % Cl and Cd from airfoil data
        Cl = Cl_fun(alpha_deg);
        Cd = Cd_fun(alpha_deg);

        % normal and tangential force coefficients

        Cn = Cl * cos(phi) + Cd * sin(phi);
        Ct = Cl * sin(phi) - Cd * cos(phi);

        % if Ct ~ 0, avoid dividing by zero
        if abs(Ct) < 1e-6
```

```
            Ct = sign(Ct) * 1e-6;
        end

        %update a and a' using standard BEM relations
        % a = 1 / (4 F sin^2(φ) / (σ' Cn) + 1)
        a_new = 1 / ( (4 * F * sin(phi)^2) / (sigma_prime * Cn) + 1 );
        a_new = min(a_new, 0.95);

        % a' = 1 / (4 F sinφ cosφ / (σ' Ct) - 1)
        denom_ap = (4 * F * sin(phi) * cos(phi)) / (sigma_prime * Ct) - 1;
        ap_new   = 1 / denom_ap;

        %for stability
        relax = 0.3;
        a  = (1 - relax) * a  + relax * a_new;
        ap = (1 - relax) * ap + relax * ap_new;

        % clamp to physical range
        a  = min(max(a,  0.0), 0.95);
        ap = min(max(ap, -1.0), 1.0);

        % convergence test
        if abs(a - a_new) < tol && abs(ap - ap_new) < tol
            break;
        end
    end

    %compute forces using the converged a and a'
    V_axial = U_inf * (1 - a);
    V_tan   = Omega * ri * (1 + ap);
    V_rel   = sqrt(V_axial^2 + V_tan^2);
    q       = 0.5 * rho * V_rel^2;

    % use the same Cl, Cd, phi, Cn, Ct at convergence
    % tangential force per unit span:
    Ft_prime = q * ci * Ct;     % [N/m]
    Ft(i)    = Ft_prime;

end

% 9) integrate torque and compute Cp
Q = B * sum(Ft .* r .* dr);          % torque [N·m]
P = Omega * Q;                       % power [W]
A = pi * R^2;                        % swept area
P_ref = 0.5 * rho * A * U_inf^3;     % reference power

Cp = P / P_ref;

end
```

### File: Run_BEM_CP_curve.m

```
clear; clc;

% load all geometry, constants, and airfoil tables
BEM_init;   % this runs the init script you wrote earlier

Cl_fun = Cl_lin;
Cd_fun = Cd_lin;

% define a range of tip-speed ratios
lambda_vec = linspace(1, 8, 21);   % for example
Cp_vec = zeros(size(lambda_vec));

% compute Cp for each λ
for k = 1:numel(lambda_vec)
    lam = lambda_vec(k);
    Cp_vec(k) = BEM_single_lambda(lam, R, B, rho, U_inf, ...
                                  r, c, theta, Cl_fun, Cd_fun);
end
```

```
% 4) plot Cp-λ curve
figure;
plot(lambda_vec, Cp_vec, 'o-','LineWidth',1.5);
xlabel('\lambda (tip-speed ratio)');
ylabel('C_P');
title('C_P vs \lambda for standard blade');
grid on;
```

### File: design_optimized_blade.m

```
% Create an optimized blade using textbook rotor theory (Manwell 3.9, 3.10.1)
% -lambda_design = 6 (typical small turbine, avoids over-optimization)
% -alpha_design chosen from Cd/Cl minimum in usable region
% -Optimal phi, twist, and chord generated
% -Manufacturability constraints enforced

clear; clc;


%Load constraints from standard blade
BEM_init;    % loads: R, rho, B, U_inf, r, c, theta, alpha_tab, Cl_tab, Cd_tab

% Wrap standard blade
blade_std.r     = r;
blade_std.c     = c;
blade_std.theta = theta;
blade_std.name  = "Standard blade";

% Choose interpolation
Cl_fun = Cl_lin;
Cd_fun = Cd_lin;

% Design tip speed ratio
lambda_design = 5;

fprintf("Design TSR (lambda_design) = %.2f\n", lambda_design);

% Choose alpha_design from airfoil Cd/Cl minimum in pre-stall range
valid = alpha_tab >= 0 & alpha_tab <= 12 & Cl_tab > 0;

alpha_use = alpha_tab(valid);
Cl_use    = Cl_tab(valid);
Cd_use    = Cd_tab(valid);

ratio = Cd_use ./ Cl_use;

[~, idx_min] = min(ratio);

alpha_design_deg = alpha_use(idx_min);
alpha_design_rad = deg2rad(alpha_design_deg);
Cl_design        = Cl_use(idx_min);

fprintf("Chosen alpha_design = %.2f deg (Cl=%.3f)\n", ...
        alpha_design_deg, Cl_design);

%Compute local lambda_r and inflow angle phi(r)
lambda_r = lambda_design * (r / R);

phi_opt = (2/3) * atan(1 ./ lambda_r);

% compute optimized twist
theta_opt = phi_opt - alpha_design_rad;
theta_opt_deg = rad2deg(theta_opt);

% Compute theoretical optimal chord
% c_theory(r) = (8*pi*r)/(B*Cl) * (sin(phi) / (lambda*(1+cos(phi))))

c_theory = (8*pi .* r ./ (B * Cl_design)) .* ...
           (sin(phi_opt) ./ (lambda_design * (1 + cos(phi_opt))));

% Make sure no crazy values
```

```
c_theory = max(c_theory, 1e-4);

c_root_theory = c_theory(1);
c_tip_theory  = c_theory(end);
c_root_des = max(c_root_theory, 0.04);  % at least 4 cm at root
c_tip_des  = max(min(c_tip_theory, 0.02), 0.01);  % tip between 1–2 cm
xi = (r - r(1)) / (r(end) - r(1));   % goes from 0 at root to 1 at tip
c_taper = c_root_des + (c_tip_des - c_root_des) .* xi;   % linear taper
mean_theory = mean(c_theory);
mean_taper  = mean(c_taper);

scale_c = mean_theory / mean_taper;
c_opt   = scale_c * c_taper;

% enforce minimum chords again
c_root_min = 0.03;   % 3 cm at root
c_tip_min  = 0.01;   % 1 cm at tip

c_opt(1)   = max(c_opt(1),   c_root_min);
c_opt(end) = max(c_opt(end), c_tip_min);

% light smoothing
c_opt = smooth(c_opt, 3)';

%manufacturability constraints

% minimum chord limits
c_root_min = 0.03;   % 3 cm at root
c_tip_min  = 0.01;   % 1 cm at tip

c_opt(1) = max(c_opt(1), c_root_min);
c_opt(end) = max(c_opt(end), c_tip_min);

% Smooth chord to avoid wild shapes
c_opt = smooth(c_opt, 3)';   % small moving window smoothing

% Package optimized blade
blade_opt.r                 = r;
blade_opt.c                 = c_opt;
blade_opt.theta             = theta_opt;
blade_opt.lambda_design     = lambda_design;
blade_opt.alpha_design_deg = alpha_design_deg;
blade_opt.name              = "Optimized blade";

% Plot chord & twist comparison
figure;
subplot(2,1,1);
plot(r, rad2deg(theta), 'o-', r, theta_opt_deg, 's-', 'LineWidth',1.2);
xlabel("r [m]");
ylabel("Twist θ [deg]");
legend("Standard", "Optimized");
title("Twist distribution");
grid on;

subplot(2,1,2);
plot(r, c, 'o-', r, c_opt, 's-', 'LineWidth',1.2);
xlabel("r [m]");
ylabel("Chord c [m]");
legend("Standard", "Optimized");
title("Chord distribution");
grid on;

% Compare CP(λ)
lambda_vec = linspace(1, 8, 25);
Cp_std = zeros(size(lambda_vec));
Cp_opt = zeros(size(lambda_vec));

for k = 1:numel(lambda_vec)
    lam = lambda_vec(k);

    Cp_std(k) = BEM_single_lambda(lam, R, B, rho, U_inf, ...
```

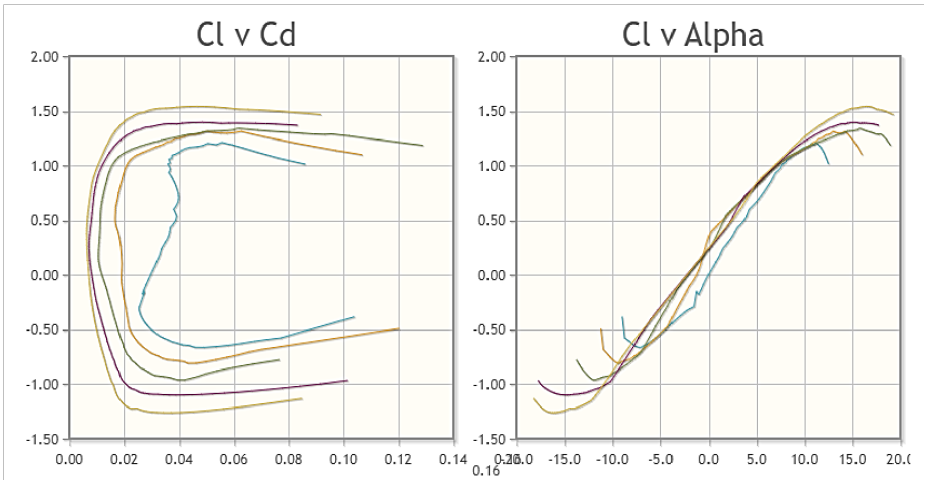
```
                                blade_std.r, blade_std.c, blade_std.theta, ...
                                Cl_fun, Cd_fun);

    Cp_opt(k) = BEM_single_lambda(lam, R, B, rho, U_inf, ...
                                blade_opt.r, blade_opt.c, blade_opt.theta, ...
                                Cl_fun, Cd_fun);
end

figure;
plot(lambda_vec, Cp_std, 'o-', lambda_vec, Cp_opt, 's-', 'LineWidth',1.2);
xlabel("\lambda");
ylabel("C_P");
legend("Standard","Optimized");
title("C_P(\lambda): Standard vs Optimized");
grid on;

fprintf("Done. blade_opt and blade_std stored in workspace.\n");
%%Package optimized blade
blade_opt.r                 = r;
blade_opt.c                 = c_opt;
blade_opt.theta             = theta_opt;
blade_opt.lambda_design     = lambda_design;
blade_opt.alpha_design_deg = alpha_design_deg;
blade_opt.name              = "Optimized blade";

% Print optimized blade geometry table

theta_opt_deg = rad2deg(theta_opt);    % twist in degrees (easier to read)

fprintf('\nOptimized blade geometry (similar format to standard blade table):\n');
fprintf('  i   r_base [m]   r_hub [m]   theta_opt [deg]   chord_opt [m]\n');
fprintf(' ----------------------------------------------------------------\n');

for i = 1:numel(r)
    fprintf('%3d   %8.4f     %8.4f      %8.3f        %8.4f\n', ...
            i, r_base(i), r(i), theta_opt_deg(i), c_opt(i));
end
fprintf(' ----------------------------------------------------------------\n\n');

% create a MATLAB table object (nice for export / copy)
geomTable = table((1:numel(r))', r_base(:), r(:), theta_opt_deg(:), c_opt(:), ...
    'VariableNames', {'Index','r_base_m','r_hub_m','theta_deg','chord_m'});

disp('Optimized blade geometry table (MATLAB table object):');
disp(geomTable);
```

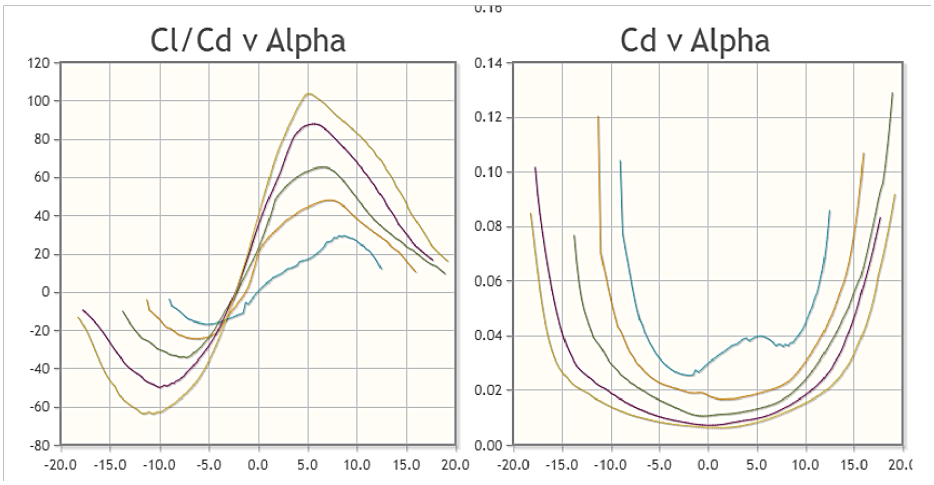
### File: compute_AEP_weibull

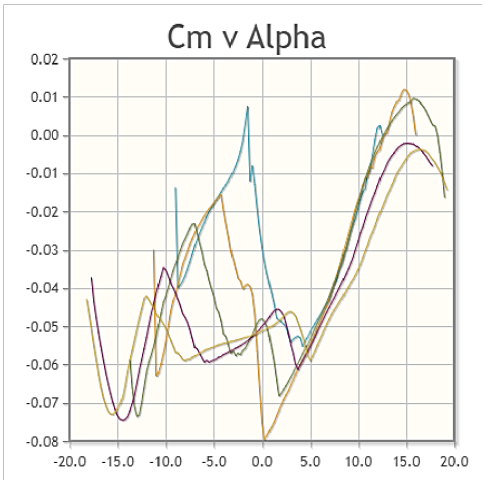
```
% Computes annual energy production (AEP) using Weibull distribution

clear; clc;

% Load blade designs and BEM system
BEM_init;        % loads R, rho, B, U_inf etc.
design_optimized_blade;   % generates blade_std and blade_opt

lambda_design = blade_opt.lambda_design;
Cl_fun = Cl_lin;
Cd_fun = Cd_lin;

%Weibull distribution parameters
U_mean = 6;      % mean wind speed [m/s]
k = 2.2;         % Weibull shape

% Compute scale parameter c
c = U_mean / gamma(1 + 1/k);

% Create wind speed vector
U = linspace(0, 25, 400);   % covers almost all probability
dU = U(2) - U(1);
```

```
% Weibull PDF
f_U = (k/c) * (U./c).^(k-1) .* exp(-(U./c).^k);

% Plot Weibull distribution
figure;
plot(U, f_U, 'LineWidth', 1.8)
xlabel("Wind speed U [m/s]")
ylabel("PDF f(U)")
title("Weibull Wind Speed Distribution (k=2.2, mean=6 m/s)")
grid on;

% Compute CP for lambda_design for both blades
Cp_std = BEM_single_lambda(lambda_design, R, B, rho, U_inf, ...
                           blade_std.r, blade_std.c, blade_std.theta, ...
                           Cl_fun, Cd_fun);

Cp_opt = BEM_single_lambda(lambda_design, R, B, rho, U_inf, ...
                           blade_opt.r, blade_opt.c, blade_opt.theta, ...
                           Cl_fun, Cd_fun);

fprintf("Cp (standard)  at TSR %.1f : %.4f\n", lambda_design, Cp_std);
fprintf("Cp (optimized) at TSR %.1f : %.4f\n", lambda_design, Cp_opt);

% Compute power vs wind speed
A = pi * R^2;

% Standard blade power curve
P_std = 0.5 * rho * A .* U.^3 * Cp_std;   % watts

% Optimized blade power curve
P_opt = 0.5 * rho * A .* U.^3 * Cp_opt;   % watts



% Compute AEP
hours_year = 24 * 365;

AEP_std_Wh = sum(P_std .* f_U * dU) * hours_year;
AEP_opt_Wh = sum(P_opt .* f_U * dU) * hours_year;

AEP_std_kWh = AEP_std_Wh / 1000;
AEP_opt_kWh = AEP_opt_Wh / 1000;

fprintf("\n===== Annual Energy Production (AEP) =====\n");
fprintf("Standard blade:   %.2f kWh/year\n", AEP_std_kWh);
fprintf("Optimized blade:  %.2f kWh/year\n", AEP_opt_kWh);
```

## 10 Appendix 3: Airfoil data NACA 2414, CD, CL as a function of AoA




## 11 Appendix 4: Data for calculations

## 12 References:

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Wind Turbine Aerodynamics and CFD Simulation—Lecture Slides, Assignments, and Reference Resources.

Provided by Dr. B. Bhaskar during Fall 2025.

(Used for background theory, BEM model formulation, CFD setup guidance, and verification of analytical assumptions.)